COHERENCE OF THE NOETHERIAN INTERSECTION F CAP F_*G UNDER A FLAT SURJECTIVE BASE CHANGE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 07UB: sources/stacks-chow/tags.txt, source lines 9966–9966. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07UD: sources/stacks-chow/tags.txt, source lines 9968–9968. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 060U: sources/stacks-chow/tags.txt, source lines 7738–7738. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 073K: sources/stacks-chow/tags.txt, source lines 9064–9064. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module, and let $\varphi : \mathcal{G} \to f^*\mathcal{F}$ be a module map. Assume*

1. *φ is injective,*
2. *f is quasi-compact, quasi-separated, flat, and surjective,*
3. *X, Y are locally Noetherian, and*
4. *$\mathcal{G}$ is a coherent $\mathcal{O}_Y$-module.*

Then $\mathcal{F} \cap f_*\mathcal{G}$ defined as the pullback

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & f_*f^*\mathcal{F} \\ \uparrow & & \uparrow \\ \mathcal{F} \cap f_*\mathcal{G} & \longrightarrow & f_*\mathcal{G} \end{array}$$

is a coherent $\mathcal{O}_X$-module.

Proof. We will freely use the characterization of coherent modules of Cohomology of Spaces, Lemma 07UB as well as the fact that coherent modules form a Serre subcategory of $QCoh(\mathcal{O}_X)$, see Cohomology of Spaces, Lemma 07UD. If f has a section σ, then we see that $\mathcal{F} \cap f_*\mathcal{G}$ is contained in the image of $\sigma^*\mathcal{G} \to \sigma^*f^*\mathcal{F} = \mathcal{F}$, hence coherent. In general, to show that $\mathcal{F} \cap f_*\mathcal{G}$ is coherent, it suffices the show

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that $f^*(\mathcal{F} \cap f_*\mathcal{G})$ is coherent (see Descent on Spaces, Lemma 060U). Since f is flat this is equal to $f^*\mathcal{F} \cap f^*f_*\mathcal{G}$. Since f is flat, quasi-compact, and quasi-separated we see $f^*f_*\mathcal{G} = p_*q^*\mathcal{G}$ where $p, q : Y \times_X Y \to Y$ are the projections, see Cohomology of Spaces, Lemma 073K. Since p has a section we win. $\square$